makes it look as though it were partly due to the structure or the quality of the steel.

John Tonge (M. & St. L.): There is one point Mr. Vaughan has not touched upon, which no doubt he will recognize, and that is in the hard tire you don't get the tonnage load over the roadway, as a rule, that you would with a softer tire on the same engine. It will make a material difference in the tonnage on the level, and more especially on the grades.

G. R. Joughins (Intercolonial): One of the most remarkable things we have noticed is the fact that we have very little trouble with driving tires. It is nearly all with either car wheel tires or tender tires, all of a small size, and I concluded from that, in the same way that Mr. Vaughan has, that it depends very much on the work which is put on the steel. Whether it is due to the much greater rolling which the driving tire receives, compared with the small tire, I don't know, but we have tried a good many makes of steel, and some of them have been very expensive. We have bought steel tires which were the highest priced in the market, so far as we could find—of course, I mean the Canadian market, and that includes tires which we get from Europe—and the results have been no better with the high priced tire than with the cheap one. We are in very serious difficulty in the matter, because in the winter time a large number of tires give us trouble and breakages, and we are a little afraid of what the results will be. In the summer time we have practically no trouble.

C. E. Fuller (U. P.): We have had the same trouble, and in discussing it with one of the large tire manufacturers, a man of a great deal of experience, he admitted that the tire for a tender wheel or car wheel was not suitable as a tire for a locomotive, and a locomotive tire was not suitable for a car wheel. The punishment that the tender wheel receives is terrific, and he intimated that the manufacturers could and would have to make a tire that would stand up under tender service. I believe that this committee should not be discharged, but should be requested to work with the manufacturers and experiment to get a tire that will stand up under tenders. We have just gone through a severe winter and have had all kinds of trouble. If the wheel men realize that they have to correct the trouble, they will correct it. They are just as anxious to correct it as we are, and I know of no better way for them to get in touch with the true situation than to do so through a committee. I make a motion that this committee be extended and instructed to work with the tire manufacturers with a view of getting a better tire made for tender and car service—one that will overcome this trouble.

C. E. Chambers (C. R. R. of N. J.): Have any of the members had any experience with vanadium tires?

H. T. Bentley (C. & N. W.): We have a set of tires on a switch engine, and from a report received the other day from the master mechanic, it appears as if there was very little, if any, difference in the wear of the vanadium steel tires as compared to other tires on the same engine, put on at the same time. The manufacturers, of course, try to tell us that the shelling out of tires is due to the locomotive department; that we are responsible for it. One man came to me and told me positively that if our wheels didn't slide we would never have any shelled out spots. He happened to come around one day when we had a very bad case of a shelled out tire on one wheel, and on the other wheel there was absolutely no indication of sliding. Have any of the members used a heavier tire than 3½ in. in thickness? If so, have they received better or worse service? We are afraid that by getting a very thick tire, like 4 or 4¼ in., the rolling would not be as good as would be possible with a thinner, or a 3½ in. tire, such as we are using.

Mr. Tonge: In the fall of 1884 we put two different tires, 5 in. thick, in passenger service, the engine running every night. One of the men that ran the engine is now in the room. There was a great difference in the service of the two tires. One set, at the last report, had run over a million miles and they were still 2¾ in. thick. The other set, when they were about one-half worn, showed cavities around the rim.

F. F. Gaines (C. of Ga.): Several years ago I applied some 4-in. tires to heavy switch engines; the results were not altogether satisfactory. The tires were unquestionably softer and wore faster, and they wore irregularly; they didn't seem to be uniform and were evidently not as well worked as the tire of less thickness. Since that time I have kept down to 3½ in.

Mr. Chambers: What class of engine were the tires applied to which Mr. Tonge mentioned? We have had in service for ten or eleven months a set of vanadium tires on a consolidation engine weighing about one hundred and eight tons. They have not been turned as yet, but I see no perceptible difference in the wear of these tires from that of the ordinary steel tire.

Mr. Tonge: Our engine weighed 40 tons, was in our Chicago train service, running from St. Paul to Albert Lea, connecting with the Rock Island.

Mr. Gilbert: Mr. Vaughan has, I think, recently introduced some passenger trucks on tenders, and I would like to ask if anyone can give any information as to whether the character of the truck has any effect on the wear of the tires. I know that we have comparatively little trouble on passenger equipment where we use passenger trucks, but since the introduction of the Diamond truck and high speed tenders, our troubles seem to have increased.

Mr. Bentley: The tire I mentioned as having been very badly honeycombed on one side, and not on the other, was under a pedestal type truck.

Mr. Gaines: Is not the chief object in using vanadium steel to ensure a tire that will not break, rather than for adding to its wearing qualities? Has anyone any information as to the relative strength of the vanadium steel tire as compared to the ordinary tire?

Mr. Chambers: I have no figures, but I would not care to pay the difference in the price of the tires to have the little assurance that when they got to a certain size they would not break. We have very little trouble with breakage of tires above the scraping limit.

Mr. Bentley: We had during this last winter several cases of 3½-in. new tires breaking a clear break, something I never saw before in my railroading experience.

Mr. Vaughan: I am not sure, but I have a feeling that there is a difference between the diamond truck and the pedestal truck in causing tires to shell out. We used the diamond truck on tenders, and as an instance of how severe the service is, on a rock bottom road bed in very cold weather we really gave up the diamond truck because we were not able to maintain it. We would chew the threads off the column bolts and even took pieces out of the oil boxes and that sort of thing. It was something awful. But the pedestal truck has given better service and those troubles have largely gone. On the division I am referring to, which is our worst division for those conditions, we have had less trouble recently with the pedestal truck with shelled out wheels, than we previously had with the diamond truck. Whether it is due to the tire having been made differently or not, or due to the truck, I am really unable to say, but I have a feeling that there is something in the amount of dead weight on the diamond truck which is pretty high.

Theoretically, and I believe practically, the tender service is the most severe service we put the tire to. Theoretically, the strain on the tire is inversely in proportion to the diameter of the wheel, with a given load. For instance, the tires on a 34 in. wheel under tender weighing 20,000 lbs., are practically strained as severely as the tires on a 69 in. wheel under a tender weighing 40,000 lbs.

We are blaming the tires for breaking, many times, when the trouble is really due to the wheel centers. We have had several cases of tires breaking during winter service, and in every case but one they have been on a light type of cast steel wheel center. If you will try shrinkage tests on the type of wheel center that was in use four or five years ago, with a light rim and spokes thinned down, you will find that it will collapse under the tire almost as much as you want. There is very little strength in the center to resist the force due to the load on the wheel. Many of the light designs of cast steel wheel centers will spring under the load of the engine in heavy winter weather and the tire will take part of the strain and bend. You cannot keep on bending a 3½ in. tire with the weight of the engine on it, without starting a crack. It is my conviction that, in the case of the thick tires which crack, the cracks are due to light designs of cast steel wheel centers. When we had the cast iron wheel center, although it was not as strong as the cast steel, it would stand up under the load, not deflect; whereas when the builder began to lighten up the cast steel wheel center, there was a chance for the bending action. There is the same chance for the bending action in a good many designs of the forged steel wheel centers. We have found that the forged steel wheel center will collapse under the tire to an almost indefinite extent. We make our cast steel wheel centers as stiff as the old cast iron centers, and believe it is the right thing to do. We have also stiffened up the tender wheels by making them of heavy steel castings, to get sufficient rigidity to hold the wheels in shape under the loads they are subjected to. We have less trouble with tire breakages, in the case of the thin tire almost worn to the scrapping line, than we have with thick tires.

Mr. Bentley: The two breakages I referred to were in the case of a new design of wheel centers, on new engines just from the builders.

Mr. Vaughan: Were they stiff centers?

Mr. Bentley: Yes, they were well-designed centers.

brasses, shoes and wedges snug, and free from pounding, there would be less trouble with frame breakage, and our rod brasses would not need renewing so often.

With this in mind, I am giving a few suggestions that may possibly enable us to keep an engine out of the shop until such time as the driving axles need renewing, unless a frame breakage occurs to take the engine out of service. To get the best results, driving boxes should be made of cast steel, and designed of such proportions in shoe and wedge fit as to give long life, and with an adjustable or removable hub liner, so that lateral motion could be taken up with wheels in place. The driving-box brasses should be made of ample size and made of suitable material for obtaining long lite, and of a removable type so that it will be possible to quickly replace them without having to drop the wheels.

Shoes and wedges should be of such a size that a large bearing surface will be in contact with the driving boxes. With the ordinary design, the shoe and wedge face is altogether too small, and the wear very rapid. Flangeless shoes and wedges should be used to overcome the trouble experienced with flanges breaking. To facilitate the lining down of wedges, arrangements should be made so that they can be removed without disturbing the binders, or underhung springs; the wedge bolts should be of sufficient strength and so arranged that in case of breakage it will be an easy matter to replace them. Binders to be so designed that they will securely hold the frame jaws together, and prevent movement, but yet of such construction that they can readily have wear taken up when necessary.

Frames.—Most roads have more or less trouble with frame breakages, and, if these could be eliminated by either improving the design or making them of some material that would stand up under the shocks they are subjected to, a step in the right direction would be made. I noticed an article some time ago on the great amount of frame breakage that was occurring with engines having the Walschaert valve gear. Upon looking into this on our own road, I found we are not having the slightest trouble; not a single case of frame breakage has occurred on any outside gear engine during the past four years.

Frame Construction for Engines with Outside Valve Gear.—As most locomotives are now being built with outside valve gear, there is very little difficulty in designing a suitable cross bracing that will add materially to the strength and life of frames.

The replies to a circular of inquiry are summed up as follows:

No. 1. Have you any suggestion to make in the way of improving the driving box now in general use, and, if so, what do you recommend? Of twenty-six answers sixteen replied "No," whereas the balance suggested using heavy steel boxes, except in one case, where it was recommended that the pedestal jaws be spread farther apart, so that heavier *cast-iron* boxes could be used.

No. 2. Have you any way of taking up lateral wear in driving boxes without removing them? Very little has been attempted in this direction, although replies indicate there is a great need of something that will enable it to be done.

No. 3. Are you using driving boxes with brasses that can be taken out without dropping wheels? Notwithstanding the desirability of such an arrangement, only five roads report using anything of the kind, and all are using the same patented device, experimentally; in three cases satisfactorily, in one case it has not been in service long enough to report on, while the fifth user did not find it entirely satisfactory, but did not state in what respect it failed.

No. 4. What mixture or special metal do you use in driving-box brasses, and is it entirely satisfactory? Do you use grease for lubricating switch engines, and is it giving satisfaction? In the first section of this question, most roads reply that they are making their own mixture (copper 80 per cent, tin 10 per cent and lead 10 per cent), with very satisfactory results, while a few are purchasing special brands, which give good service. In answer to second part of the question there appears to have been a difficulty experienced while using grease in switch engines, and it early became evident that the kind of grease and perforated plates working satisfactorily on road engines would not answer the purpose for the slower moving switch engine, and, therefore, a thinner grease and perforated plates with larger holes were introduced, and this combination appears to give better results.

No. 5. Do you use adjustable or solid shoes and wedges, and are they satisfactory? The general practice appears to favor the solid shoe with adjustable wedge, and, as a whole, it is satisfactory, although with limited bearing surface considerable wear takes place. With engines having Walschaert or other outside gear, there is no reason why the frame jaws cannot be designed to get a width of eight to ten inches, if desired, so that the pressure per square inch could be greatly reduced. On a large number of European engines a solid pedestal of great width is used, with very satisfactory results.

No. 6. Do you have much shoe and wedge flange breakage? If so, how do you overcome it? Considerable trouble appears to have been experienced in this direction and has been overcome in some cases by using bronze shoes and wedges; in others, by thickening flanges where possible, and, on most roads, frame jaws are now rounded off so that a good fillet can be left in shoes and wedges. We have entirely overcome the breakage of flanges by simply leaving them off, and using side plates riveted on the frame, the flanges of the driving box coming in contact with these side plates, instead of the flanges of shoes and wedges, as formerly.

No. 7. What width of bearing face do you have on shoes and wedges where they come in contact with driving boxes? The replies indicate that this varies on different engines and roads, the minimum being 4 in. with 7¾ in. as a maximum, depending on the size of engine. With consolidation engines in service on the Chicago & North Western we have a wedge face of 8¼ by 17½ in., with a pressure per square inch of 122.8 pounds, as compared with our former standard freight engine with 6¼ by 17 in. wedge face, with a pressure of 184.72 pounds per square inch.

No. 8. With outside valve-gear engines, have you tried to increase the width of shoe and wedge face, and, if so, how? In no case does there appear to have been an attempt made to increase frame jaws on engines having outside valve gear, and yet it is a simple proposition, and by doing so increased wearing surface could be obtained.

No. 9. Have you any way of taking down wedges without removing binders or driving box? In no case has this been attempted, although it can be accomplished by using flangeless wedges and cutting away the inside flange of the driving box on the wedge side; what was formerly a four or five-hour job, depending upon size of engine, can be accomplished in less than an hour.

No. 10. Do you use brass or cast-iron shoes and wedges? The general practice appears to favor cast iron for this purpose, but a number of roads prefer bronze shoes and wedges where steel boxes are used, while a few people seem to like a bronze liner on a steel box, and then use a cast-iron shoe and wedge. With this latter arrangement, we have not had very good results, account of difficulty in keeping liners fast on the box; our present practice is to use cast-iron shoes and wedges against steel boxes in freight service, and bronze bearing on passenger engines.

No. 11. Can you line down shoes and wedges without taking them out to apply liners, and, if so, how? The replies are practically the same as to question 9, except that occasionally loose liners are inserted behind the wedge, which, however, can only be done by taking the binder down, so that what should be a simple job is a difficult and expensive one.

No. 12. With underhung springs can you remove and replace broken wedge bolts without taking binders down? Where underhung springs are used, it seems impossible on most roads to remove or apply wedge bolts without removing springs. We had a similar difficulty, and practically overcame it by making binders with slotted holes, which enabled us to take out and replace wedge bolts with springs in place, thus reducing the job from a big to a small one.

No. 13. Which type of binder do you find the most satisfactory? The pedestal cap type meets with most favor, as replies from fourteen roads indicate it is their preference, and that frame breakage is reduced where it is used. The strap binder is next best, seven replies being in favor of it. The clamp over the frame jaw lugs is used on four roads on account of simplicity, and only one road is in favor of using the thimble and bolt. My personal preference is for the pedestal cap type.

No. 14. Have you anything to suggest in the way of a binder that will take up wear without having to be upset and refitted? The replies indicate that the pedestal cap type binder does enable the wear to be taken up with less trouble and expense than any other make, in which conclusion I concur.

No. 15. What suggestion have you to offer in regard to frame construction on engines with outside valve gear? The proper cross bracing of frames is the most logical use to make of the space that was formerly taken up by valve gear, and will do more to overcome frame breakage than almost anything else. One suggestion is to use upper and lower rails over the cylinder casting. This design can and is used, however, with engines not having outside valve gear, and is a great help in reducing front frame breakages and loose cylinders.

No. 16. Do you have as much trouble with frame breakage

you had a key there, and did not chip it off even with the frame, but let it project so that a man could give the key a blow occasionally, and keep it home, you have a better support. There is no job around the engine that deserves more attention than the tail piece at the back end of the frame. The better cross-bracing we can get, the safer we are and the better satisfaction we have with the frames with regard to loosening. We want our driving box brasses to have as much bearing as possible. Furthermore, the wedge must not be allowed to get loose or be down. Keep it up and in proper working order. I believe with these matters attended to, and what has been brought to our notice, we ought to prevent our frames from breaking.

C. E. Fuller: What kind of a side plate does Mr. Bentley rivet to the frame, where he uses the flangleless shoes and wedges; how does he fasten it?

The President: Mr. Chambers, you have been using the one-piece frames for some time. What service are you getting with these frames?

C. E. Chambers (C. of N. J.): We have used locomotives with one-piece frames for several years. Some of them have a front section which is bolted on under the cylinders, so that in case of accident, such as some of the speakers have mentioned, it would only be necessary to renew the front section. We also have other engines with the one-piece frame. If any breakage occurs on the frame back of the front end, it is welded by thermit. We have several of them which have been in successful use for a number of years. The one-piece frame with the removable section in front is a very satisfactory frame.

C. A. Seley (C. R. I. & P.): There is one point in connection with Mr. Gaines' advocacy of cast iron, and that is the question whether it is a better wearing metal than cast steel —cast steel is very bad, and we have all felt the necessity of covering up the surface with something that will protect it. In the west we have used babbitt inserts and brass linings to a considerable extent. I am inclined to believe, where we are under the necessity of using cast steel boxes, that the babbitt inserts at times may practically render the frame satisfactory.

Mr. Bentley: Have any of the members tried some kind of removable brass? I understand on some roads they have a brass similar to a car brass or tender-truck brass, probably on the front and back boxes, which they can take out by jacking the wheel up a little. I think no one has yet had the courage to put a removable box in the main bearing.

F. F. Gaines (C. of Ga.): I was formerly connected with a road where that practice was in vogue on everything except the main box. It was made similar to a tender brass, and you would have to jack it up sufficiently to get a 1½ in. clearance so that you could slide the brass out. We never put it on the main box, because I do not think it would work there. There is too much work and too much motion to put it on the main box, but it worked very satisfactorily on the other boxes.

Mr. Bentley: We have had a number of engines equipped with removable driving box brasses for a good many years, and I have been personally following it up where I could. Some of the savings we have made by the use of this removable brass are remarkable. The fact that you can take an engine into the roundhouse in the morning and by noon have a new brass in it, and have it on the road in the afternoon, as compared with having it out of service for one or two days, which was formerly our practice, is most important. I have some figures that show the cost of driving box brasses, shoes and wedges, and lateral motion plates. We have a lateral motion plate that we can put in position with the wheels under the engine; it may be interesting to know how far we have gone in this direction. We hope it will be the means of everybody doing likewise, as all the roads would gain considerable advantage from such a practice. We find it costs a little more in the first place to apply flangeless shoes and wedges than it costs in the case of the ordinary cast iron shoes and wedges, the difference being $6.49 in favor of the ordinary cast iron ones. Where you have to line down the cast iron shoes and wedges with flanges, as compared with removing the flangeless shoes and wedges, the saving is something remarkable. I have frequently seen engines fitted with a cast iron flanged shoe and wedge have the flanges broken off going around some of the sharp curves entering the Wells Street Depot the day after they were put on, and we had to put in new wedges. With the flangeless wedges this is unnecessary.

In the use of the driving box brass, we can save probably $30, considering the time the engine is out of service, by putting in the removable brass, as compared with dropping the wheels under the old practice and taking the box to the machine and putting the new brass in and getting it back on the engine.

J. A. Pilcher (N. & W.): There has not been any epidemic of breakages, and we have been led to the conclusion that it is better policy for us to adhere to the cast iron boxes, making the box frames very deep to get the strength.

D. R. MacBain: The New York Central Lines some years ago adopted a method of lining steel driving boxes with brass, for the purpose of getting away from the excessive box and shoe and wedge cutting. It was a considerable item of expense and a couple of years ago a committee was appointed to make a series of experiments with cast iron, with a view of determining whether or not we could use cast iron on modern power. The committee has arranged, as I understand the matter, to have about twenty consolidation engines equipped with high grade cast iron boxes, of two different mixtures. Some of the engines have been in service a year, and up to the time I left the New York Central we had not had any failures. The operation of the box so far has been very satisfactory. We simply took out the cast steel box and put in the cast iron boxes with the same dimensions of shoe and wedge.

Mr. Bentley: As I understand Mr. MacBain, he was using a cast iron box instead of a steel box with a cast iron shoe and wedge.

Mr. MacBain: We were using the cast steel box and we started to experiment with the cast iron box.

Mr. Bentley: Mr. MacBain brought up the question of increasing the length of the box and brass, but he seemed to have overlooked what I consider the most vital point in the whole thing, and that is increasing the jaw dimensions. The locomotive builders do not like that kind of an arrangement, because it means a lot of work for them, but if you can increase your jaw fit you will get a greatly increased life from the shoes and wedges.

Mr. Gaines asked about the driving axle. I was reaching rather high probably in trying to make the engines stay on the wheels until the axle was worn out. I saw a driving axle the other day that had only been in service two years, and I do not think that is a long life for a driving axle. It had to be taken out and refitted.

Mr. Fuller spoke about my saying that they put the brass up to the boiler. I did not say that—that was a statement made by one of the gentlemen who replied to my circular of inquiry. We run our brasses up to the boiler over the frame, and instead of riveting them or fastening them on with studs, which causes considerable trouble, we let the boiler rest in an angle iron or T-iron.

Mr. DeVoy spoke about an engine having outside valve gear—he could not see how that would affect the breakage of frames. I did not claim the kind of valve gear we were putting on our engines was responsible for the absence of frame breakages. What I do claim is that the possibilities for cross braces on the outside geared engine is the thing that has helped us.

Mr. Pratt quoted me as saying something that I did not say, and that I did not mean. I did not say anything about a one-piece frame. We have a large number and they are very nice when nothing happens and they do not get into a wreck, but when it becomes necessary to make repairs to one end of a one-piece frame, it is a big job, because you cannot repair the frame in place. I am in favor of a slab connection, where possible, so as to reduce the size of the frame for convenience of repairs.

Mr. Brown spoke of the care given to frames. I understand if you keep the pounding out of the driving box and keep the shoes and wedges snug, that it is giving care to the frames and I do not see that you can give very much further care to them while the engines are in service.

In reply to Mr. Fuller's question. We use a mild steel plate, I think it is about three-quarters of an inch thick and about two inches wide, which we rivet on to the side of the frame, which takes the place of the flanges on the shoes and wedges. The idea of that is whatever wear is caused from the driving box is transmitted to this piece of mild steel, which is easily removed when it is necessary to replace it, and instead of having to repair the frames, we simply take the worn piece of mild steel off and put on another one.

SAFETY VALVES.

The remarks and suggestions below relate to locomotive boilers only. Further, they relate only to locomotive boilers using coal as the fuel, and under the conditions now prevailing for the stimulation of the draft by the use of exhaust steam from cylinders of the locomotive and by means of the ordinary steam blower. A series of tests were made for the committee by E. D. Nelson, engineer of tests of the Pennsylvania, to determine the maximum or worst condition that

livered back to the engine at 9 o'clock. The braces were then refitted to the frames and nine new bolts fitted by 11 a. m. The last pedestal was down and the engine raised and the wheels rolled from under at exactly 8.25 a. m.

The driving boxes were then marked and removed and the first driving wheels were placed in the lathe at 8.45 a. m., and the last tire turned and wheels delivered to the erecting shop at 3 p.m. In the meantime both main pins had been removed from the wheel centers and two new ones were fitted and pressed in and riveted over, also one new No. 2 crank pin removed and a new one fitted and applied. The right No. 1 and left No. 4 flanges were worn so thin that it was impossible to turn the tires again, so they were removed and two second-hand tires were applied and turned to size. One driving journal was returned. In addition to the new crank pins, three new crank pin collars were turned, polished and applied, 5 old collars trued up in the lathe and polished, and new nuts were turned and fitted to the new main pins. This was all completed and delivered to the erecting shop by 3 p.m., the work being thoroughly inspected and pronounced good. The engine frames were cleaned, white-leaded and tested for cracks by 9.30 a.m. The bolt gang removed 50 frame and pedestal bolts and new ones were fitted and applied, this work being completed by 5 p.m.

The driving boxes were removed from the journals immediately after the wheels were removed from the engine, the first one being delivered to the machine department at 8.30 a. m., and the last one at 8.50 a. m. Upon inspection it was found that 7 new crown brasses were required and also new brass hub liners on all the boxes. The new brasses were procured from the store department, turned, planed, pressed in, drilled and dowelled, old babbitt cut out from the side of the boxes and the boxes delivered at the fire to be poured with brass for lateral at 12 m. The heat was ready at 12.25 and the work of pouring was completed at 12.30 p.m All driving boxes were then replaned on the shoe and wedge faces, the first one being completed at 1.30 and the last one at 3 p. m.

Boxes were then rebored to fit the journal and faced for lateral, grooved and cellars refitted, and all but the main ones tinned on journal bearings, the first one being completed and delivered to the erecting shop at 2.30 p.m., and the last one at 5.30 p.m. On account of having to be spotted to the journal the main boxes were finished first, both of these being delivered at 2.30 p.m. They were immediately spotted and the grease cellars fitted and applied by 3.30 p. m. The last box was placed on the journals and completed at 5.40 p.m., and at that time the driving wheels were placed under the engine.

In the meantime the engine was squared, the jaws filed up and the pedestals refitted and box centers marked on frames, the old shoes and wedges were scraped and a complete new set was fitted up and applied; new wedge bolts and standard wedge bolt brackets were fitted to them. The first box size was obtained at 1.20 p.m., while the box was still on the planer. The men could wait no longer, and the first shoe and wedge was laid off and delivered to the machine to be planed at 2.30 p.m., this work all being completed and the full set delivered back to the engine at 5.30 p.m.

The wheeling operation was completed, wheels trammed and reported ready for main and side rods at 10 p.m. The first main rod was removed and delivered to the rod department for repairs at 7.30 and the last piece of the side rods at 8.40 a.m. The rod bolts and brasses were found upon inspection to be in very bad shape. It was necessary to ream the back end and main rods and straps and fit new bolts and also fit new bolts to the main connection spade straps, both sides, and to fit and apply new brasses and bushings all around. The work was all completed and the rods delivered back to the engine at 4 p.m., and at 12 o'clock, midnight, they were all applied ready for service and the last man in the fitting department was through with the engine.

While this work was pushed rapidly along, the same good progress was being made in other departments of the shop. For instance, the steam chests and valves were completely removed and all parts delivered for repairs by 7.30 a.m., and before 8 o'clock the valves were returned to the engine and at 8.10, one hour and ten minutes from commencing the work of stripping, the valve yokes also were received at the engine, having been inspected, examined for valve fit and the stems trued up. The valves and valve strips had been replaned and were ready for the re-spotting. Both valve seats on the cylinders were found to be in bad condition and they were refaced and the valves spotted. All steam chest studs were removed and annealed; complete new gaskets were applied to the steam chests and the pressure plates were re-lined and planed.

The valve rods were received back at the engine at 9 o'clock, having been fitted up with new bushings on the back end, one pin refitted and one new pin turned and fitted complete. The whole steam chest job, comprising valves, valve rods and glands, relief valves, steam chests, covers and casings, cylinder ports blown out with air and cleaned and port openings marked on valve stems, was completed in every detail by 4 o'clock in the afternoon, at which time the men engaged on that work went to work on another engine.

The air equipment and driver brake rigging were removed from the engine at the earliest possible moment. The air equipment, such as engineer's brake valve, double check valve, straight air valve, air pump, air gauge, triple valve, governors, straight air reducing valve and safety valve were delivered in haste, as though they were perishable goods, to the air brake department by 7.30 a.m., and all parts except the air pump were replaced on the engine by 1 p.m., having been thoroughly cleaned, overhauled and tested at the test rack. The driver brake rigging was also ably taken care of by the regularly assigned men who removed all parts in time to allow the wheels to come out before 8 a.m., the remainder being stripped by 9 o'clock. This work was given a complete overhauling, including reaming and rebushing of six of the hangers, removing old and fitting and applying new hanger posts or pins (one of which had to be drilled out of the bracket, it being impossible to remove it otherwise). New brake shoes were keyed on and fitted all around, and new brake pins were turned and applied complete and all lost motion was taken up in connections. New ends were welded on the brake beams and returned; the slack adjusting screws were removed, greased and reapplied and the brake cylinders tested.

The spring rigging was taken care of immediately after the drivers were removed, and at 8.30, just five minutes after the wheels were rolled from under, and before the engine was let down upon the blocks, all the rigging was removed with the exception of four equalizers, which, however, were removed half an hour later.

At 8.35 all spring hangers were delivered to the blacksmith shop for repairs, such as holes plugged and gib holes closed to fit the gibs. At 8.45 three spring saddles and eight driving box equalizers were delivered to the blacksmith to be welded out on box fit, the equalizers to have the holes plugged in addition. The blacksmiths turned out these jobs in an incredibly short space of time, and the pieces, one after the other, hot from the fire, were turned over to the machine side.

Here, not to be outdone by the blacksmiths, the machinist quickly completed his work and the parts were delivered back to the engine to be applied. The result was that the last piece of spring rigging was completed ready to apply by 2.30 p.m. By this time much of the rigging was already applied and immediately upon finishing the laying out of the shoes and wedges and getting them out of the way, the remaining pieces were put up so quickly that when the painter came around with his automatic spraying machine, just about 3 p.m., he found that all was O. K., on the left side and he "sprayed her black all over, inside and out." Just 40 minutes later, at 3.40 p.m., the other side was treated to a similar dose of "black fluid," and the engine was ready for her wheels.